

Consider a statement that a professor might make:

"If you get 100% on the final, then you will get an A"

If you manage to get 100% on the final, then you would expect to receive an A. If you do not get 100%, you may or may not receive an A depending on other factors. However, if you do get 100%, but the professor does not give you an A, you will feel cheated.

Note: "p only if q" says that p cannot be true when q is not true. That is, the statement is false if p is true, but q is false. When p is false, q may be either true or false, because the statement says nothing about the truth value of q.

Note: To see why "q is necessary for p" is equivalent to "if p, then q", observe that "q is necessary for p" means that p cannot be true unless q is true, or that if q is false, then p is false. This is the same as saying that if p is true, then q is true.

Example 10: Let p be the statement "Maria learns discrete mathematics" and q the statement "Maria will find a good job". Express the statement $p \rightarrow q$ as a statement in English.

Soln: If Maria learns discrete mathematics, then she will find a good job.

Note: q unless $\neg p$ means that if $\neg p$ is false, then q must be true. That is, the statement "q unless $\neg p$ " is false when p is true but q is false, but it is true otherwise. Consequently, "q unless $\neg p$ " and $p \rightarrow q$ always have the same truth value.

Converse, Contrapositive and Inverse

- The proposition $q \rightarrow p$ is called the **converse** of $p \rightarrow q$.
- The **contrapositive** of $p \rightarrow q$ is the proposition $\neg q \rightarrow \neg p$.
- The proposition $\neg p \rightarrow \neg q$ is called the **inverse** of $p \rightarrow q$.